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Week 1: Interim Deliverable
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Research Methodology Note

Papers were identified using PubMed. ChatGPT was used to help extract the Q1–Q4 structure from each abstract. After confirming each paper was relevant to the research focus, the full text was reviewed and summaries refined. A gap statement was then added to link the authors' stated limitations to the Caribbean emergency department context. Two distinct research gaps were identified from two articles, in line with the assignment requirements. The reference list was manually produced using the Vancouver format and example used in Tutorial 1. The draft 90-word problem statement is included after the paper summaries and reference list.

Paper Summaries

Paper 1 - Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review

Researchers examined whether machine learning models improve clinical outcomes and triage accuracy in emergency departments compared to traditional five-level triage systems that rely on nurses' subjective judgment. Using a scoping review based on Joanna Briggs Institute methodology, the researchers searched EMBASE, Ovid MEDLINE, and Web of Science for US-based, peer-reviewed primary research studies published between 2013 and 2023. After screening 1,140 citations, 29 studies were included in the final review. Overall, the findings showed that machine learning models consistently outperformed conventional triage systems in areas such as predictive accuracy, disease identification, risk assessment, resource allocation and prediction of length of stay. However, the authors noted that the review was limited to studies published in the United States, which may reduce how well the findings apply to healthcare systems with different staffing structures, patient populations, and resource availability. This represents a direct gap for Mercer General Hospital, a Caribbean emergency department with a distinct patient mix and resource constraints not included in any of the 29 reviewed studies.

Paper 2 - AI-driven triage in emergency departments: A review of benefits, challenges, and future directions

Researchers explored whether artificial intelligence–driven triage systems can improve emergency department patient prioritisation compared to traditional triage methods that rely on subjective clinical judgment, particularly during periods of overcrowding and mass-casualty events. To investigate this, the authors conducted a narrative review of peer-reviewed literature published between 2014 and 2024, using databases including PubMed, Scopus, IEEE Xplore, and Google Scholar. The findings suggested that machine learning triage systems can significantly improve patient prioritisation, reduce wait times, and optimise resource allocation within emergency departments. However, the author notes that the narrative review design limits quantitative assessment of evidence strength, and may exclude relevant older, non-English, and grey literature. They also highlight the potential for publication bias and a lack of consideration for region-specific differences in healthcare systems and infrastructure. As a result, the generalisability of findings to diverse settings, including Caribbean emergency departments, remains limited.

Paper 3 - TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers

Researchers investigated whether combining machine learning (ML) models could predict emergency department triage levels more accurately than a single model, with the aim of helping clinicians prioritise patients more effectively during periods of overcrowding. To test this, they developed and compared seven ML models (Support Vector Machine, Random Forest, Artificial Neural Network, Gradient Boosting Machine, Logistic Regression, XGBoost, and a stacking ensemble model that combined the three best-performing algorithms) using a Korean emergency department dataset labelled according to the Korean Triage and Acuity Scale (KTAS). The results showed that the combined model, TriageIntelli, achieved the highest predictive accuracy among all seven models, demonstrating that combining the outputs of multiple well-performing algorithms improved triage classification performance compared to relying on a single model alone. However, the authors noted that the model was developed and tested only on Korean KTAS data and had not been validated in non-Korean populations. Differences in patient demographics, clinical practices, and documentation systems limit its generalisability to other healthcare settings.

Gap 1: Araouchi and Adda (2024) developed TriageIntelli, a machine learning model trained on Korean emergency department data that outperformed six other triage prediction models. However, it was only validated on Korean patients and may not generalize to Mercer General Hospital, which uses a different triage system and serves a Caribbean patient population. A 12-week pilot study could retrain the model using Mercer triage data and evaluate its ability to reduce discrepancy between triage nurses.

Paper 4 - Safety and accuracy of AI in triaging patients in the emergency department

Researchers investigated whether a general-purpose artificial intelligence (AI) model (ChatGPT-4) could safely and accurately triage emergency department patients, at a level comparable to experienced physicians using the Canadian Triage and Acuity Scale (CTAS). Through a prospective observational study, the researchers evaluated 138 patients at King Saud Medical City Emergency Department in Riyadh, Saudi Arabia. Doctors first assigned each patient a triage score using the CTAS system, and then the same patient information was submitted to ChatGPT for independent assessment. The results showed that ChatGPT agreed with emergency physicians 85.6% of the time, demonstrating substantial agreement (Cohen's kappa $\kappa = 0.78$). However, when compared to the consultant gold standard, the model's accuracy dropped to 42.9%, with the model consistently overestimating patient severity in critical cases. The author notes that the study's main limitations included its small sample size of 138 patients and its use of a general-purpose AI model that was not specifically trained for clinical triage. Additionally, the model was not validated in healthcare systems with different patient populations and clinical workflows, such as

Caribbean emergency departments, leaving uncertainty about its safety and generalisability in real-world settings.

Paper 5 - Comparing Machine Learning and Nurse Predictions for Hospital Admissions in a Multisite Emergency Care System

Researchers examined whether a machine learning (ML) model trained on large-scale electronic health record (EHR) data could predict hospital admissions more accurately than triage nurses' clinical judgment in emergency departments. The researchers developed an ensemble ML model (XGBoost + BioClinical BERT) and trained it on 1.8 million historical emergency department visits from the Mount Sinai Health System between 2019 and 2023. The model was tested on 46,912 new emergency department patient visits in 2024 and compared directly with nurse predictions. Results showed that the machine learning model achieved higher accuracy (85.4% vs. 81.6%) and sensitivity (70.8% vs. 64.8%) than nurses. The study was conducted in a large, well-resourced multi-site US health system with highly standardised electronic health record documentation. As a result, the model's performance in smaller, resource-limited, or non-U.S. emergency departments remains uncertain.

Gap 2: It was found that a machine learning model could predict hospital admissions more accurately than triage nurses at the Mount Sinai Health System in the United States. However, the model has not been tested in smaller non-US hospitals, including some Caribbean hospitals, where data may be less structured, datasets smaller and less digitised, and computing resources more limited. It is still unclear whether a machine learning model trained on a smaller Caribbean dataset could achieve similar performance. A 12-week pilot study could train a model using Mercer's existing triage records and evaluate its predictions alongside routine nurse assessments.

Paper 6 - Sepsis Prediction at Emergency Department Triage Using Natural Language Processing: Retrospective Cohort Study

Researchers investigated whether a machine learning model could predict sepsis at the time of emergency department triage using nursing triage notes and clinical information. The model was trained and tested on over one million adult emergency department encounters collected from four academically affiliated US hospitals between 2015 and 2021. Results showed strong overall predictive performance (AUC = 0.94). Among septic patients, the model identified 36.1% of cases three hours before the first intravenous antibiotic order, 49.9% at two hours, and 68.3% at one hour. However, the model was developed and validated exclusively in US academic hospitals with detailed nursing documentation, making its performance in settings such as Caribbean emergency departments uncertain.

Paper 7 - Early prediction of sepsis in intensive care patients using the machine learning algorithm NAVOY® Sepsis, a prospective randomized clinical validation study

Researchers investigated whether the machine learning algorithm NAVOY® Sepsis could accurately predict the development of sepsis in intensive care unit (ICU) patients before clinical onset. The study prospectively evaluated 304 adult ICU patients at Skåne University Hospital in Sweden using routinely collected vital signs, blood gas measurements, and laboratory data. Results showed that NAVOY® Sepsis could predict sepsis approximately three hours before onset with good performance (accuracy of 79%, sensitivity of 80%, and specificity of 78%). However, the authors noted that the study was conducted in a single ICU and delays in manual data entry during periods of high COVID-19 workload affected the timing of alerts, limiting their usefulness for clinical decision-making.

Paper 8 - External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients

Researchers evaluated the Epic Sepsis Model (ESM), a proprietary machine learning tool used in US hospitals, by testing it on electronic health record data from 27,697 patients (38,455 hospitalisations) at Michigan Medicine, a different healthcare system from the one in which the model was originally developed. In this retrospective external validation study, the researchers evaluated how accurately the model predicted sepsis, how many cases it detected, and how often it generated alerts using scores calculated every 15 minutes. The results showed poor performance, with the model demonstrating limited ability to distinguish sepsis from non-sepsis cases (AUROC = 0.63), missing 67% of sepsis cases while generating a high volume of alerts.

References:

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3. Araouchi Z, Adda M. TriageIntelli: AI-assisted multimodal triage system for health centers. Procedia Comput Sci. 2024;251:430–437.
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Draft 90-Word Problem

Sepsis is a severe reaction to infection responsible for approximately 11 million deaths worldwide each year (Rudd et al., 2020). In Mercer General Hospital's Emergency Department, triage nurses use their clinical experience to identify patients who may be developing sepsis, but delays in treatment increase mortality risk by 4–9% for each hour of delay (AAMC, 2023). AI tools such as NAVOY® Sepsis have shown promise in identifying sepsis earlier in ICU settings in Northern Europe but have not been tested in Caribbean emergency departments (Persson et al., 2024). I propose a 12-week pilot study to determine whether a machine learning system trained on Mercer's patient records can detect at least 50% of sepsis cases at least two hours before they are diagnosed.

References:

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